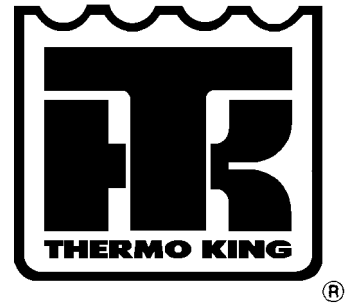




SERVICE BULLETIN



SUBJECT: μP-D Controller Replacement

BULLETIN NO. C 037

DATE: December 3, 1999

A replacement controller contains the most current software version and default settings that allow it to function when installed in a unit. However, CORRECT unit operation **REQUIRES** that the Unit Configuration, Customer Configuration and programmable Sensor Grades be set to the UNIT specific features and sensors.

NOTE: *When alarm code 42 is present, unit operating features are limited but the unit will operate in the Cool and Heat modes to control the container temperature. Alarm code 42 indicates that the Unit Configuration and Customer Configuration must still be set to activate UNIT specific features.*

μP-D Controller Replacement Guidelines

1. Determine and record correct controller configuration and sensor grade information:
 - Unit Configuration: CFG U _____
 - Customer Configuration: CFG C _____
 - RET G _____
 - SUP G _____
 - ECT G _____
2. Replace the controller.
3. Program controller setup information:
 - Unit and customer configuration settings
 - Sensor grade settings
 - Controller date and time (GMT)
 - Container identification number
 - Unit serial number
 - Compressor and On Time hourmeters
4. Perform unit Pretrip test before returning unit to service.

Step 1: Determine Correct Controller Configuration and Sensor Grade Information

To obtain the correct unit configuration, customer configuration and sensor grade information for a unit:

- a. See the Controller Setup Decal located inside the control box door. This decal will contain the most accurate setup information for the unit.

CONTROLLER SETUP INFORMATION - DO NOT REMOVE			
DESCRIPTION	DISPLAY	SETTING	CHANGES
Software Revision	REV		
Unit Configuration	CFG U		
Custom Configuration	CFG C		
Supply Sensor Grade	SUP G		
Return Sensor Grade	RET G		
Evap Sensor Grade	ECT G		
Sensors Grades of 0 have been Ice Water Calibrated.			

- b. If the Controller Setup Decal is missing from the control box, recover the Unit Configuration, Customer Configuration and programmable Sensor Grades from the previous controller.

NOTE: *See the Controller chapter in the unit maintenance manual for instructions on:*

- **Reviewing Software Version and Configuration.**
- **Menu View Functions to view GRADE submenu and hourmeter readings.**

- c. If this information can not be recovered from the previous controller (and the Controller Setup Decal is missing):
 - See Container Service Bulletin C-038 to obtain the correct unit configuration and customer configuration settings.
 - Remove the return air, supply air and evaporator coil sensors from their probe holders and read the grade stamped on the side of the sensor to obtain the correct sensor grades.

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Step 2: Replace the Controller

- a. Disconnect the unit power cord from the power supply.
- b. Disconnect all wire harness connectors from the back of the controller.
- c. Remove the screws that secure the controller to the control box door.
- d. Remove the controller and gasket from the door. Place the old controller in the anti-static bag.
- e. Install the replacement controller and gasket in the door using the existing hardware.

CAUTION: *Do NOT over tighten the mounting hardware or the controller may be damaged.*

- f. Attach the wire harness connectors to the back of the controller.

NOTE: *Be certain that all connectors are fully seated until the connector rings lock.*

- g. Plug the unit power cord into the proper power supply.
- h. Turn the unit On/Off switch ON. The unit should start and the Standard Display should appear on the controller.

NOTE: *Alarm code 42 will be present in the display when the ALARM key is pressed. This indicates that the correct Unit Configuration and Customer Configuration must still be set.*

Step 3: Program Controller Setup Information

Proceed immediately to enter controller settings. The following information **MUST** be set:

- Unit and Customer Configuration Numbers
- Sensor Grades
- Date and Time (GMT)
- Container Identification Number
- Unit Serial Number

Setting the following information is optional:

- Compressor and On Time Hourmeters

NOTE: *See the Controller chapter of the unit maintenance manual for instructions on:*

- *Menu Guard Functions to set the Unit and Customer Configuration Numbers, Sensors Grades, Time and Date, Container Identification Number, Unit Serial Number, Compressor Hourmeter and On Time Hourmeter.*

Step 4: Perform a Unit Pretrip Test

Perform a Pretrip Test to verify correct unit operation.